DWT-DCT Watermark Design Under AI-Based Image Enhancement

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***Abstract*—Classical DWT-DCT image watermarking has traditionally been evaluated against compression, filtering, and noise-based attacks, where watermark degradation is closely coupled with visible image distortion. We investigate whether AI-based image enhancement introduces a fundamentally different robustness challenge. Across 100 images from the Kodak and TAMPERE17 datasets at four embedding strengths, Real-ESRGAN reduces mean normalized correlation (NC) from 0.999 to 0.795 while maintaining the highest perceptual quality among all evaluated attacks (SSIM 0.906). In contrast, ESPCN, despite using the same $4\times$ super-resolution and Lanczos downsampling pipeline, preserves watermark recovery (NC 0.999), indicating that degradation arises from the learned reconstruction process rather than resampling alone. Coefficient-level analysis across 102,400 DCT blocks reveals markedly different perturbation profiles for Real-ESRGAN and JPEG compression, with strongly opposing coefficient rankings (Spearman $\rho \approx -0.91$). Guided by these measurements, a structured coefficient search identifies alternative embedding locations that improve Real-ESRGAN robustness while reducing JPEG robustness, demonstrating that coefficient selection becomes attack-dependent under AI-based enhancement. These findings suggest that AI-based image enhancement represents a distinct robustness evaluation scenario for classical watermarking and motivate attack-aware coefficient selection rather than fixed embedding strategies optimized for traditional signal-processing attacks.**

***Index Terms*—digital watermarking, DWT-DCT, image forensics, robustness evaluation, super-resolution, AI enhancement, Real-ESRGAN**

I. Introduction

Robustness evaluation for classical image watermarking rests on a quality–attack tradeoff: the standard benchmark attacks — JPEG compression, additive noise, filtering, geometric distortion [1], [2] — all degrade image quality in proportion to the damage they inflict on the watermark. An attacker who wishes to remove a watermark must accept perceptible image damage. This assumption is not incidental; it directly informs how embedding positions and strengths are chosen in deployed systems, and it defines what "robust" has meant in three decades of evaluation practice.

This paper asks a single question: *does consumer AI-based image enhancement violate this assumption for classical DWT-DCT watermarking — and if so, why, and what can be done about it within the classical scheme?*

The question is practically urgent. Combined DWT-DCT watermarking [3] remains a canonical choice in cameras, broadcast equipment, and digital asset management, valued for computational simplicity and independence from machine-learning infrastructure, and it continues to serve as the classical baseline in watermarking research [4]. Meanwhile, super-resolution models such as Real-ESRGAN [5] are embedded in consumer photo tools. A user who "enhances" a watermarked photograph may degrade the embedded signal with no forensic intent and — critically — no visible indication that anything was altered. Whether this constitutes a meaningful threat to deployed classical schemes has not been systematically studied: the WAVES benchmark [4] evaluated DWT-DCT only in an appendix and included no enhancement-class attacks, and W-Bench [6] evaluates only neural watermarks.

We answer the question in three stages, which structure the paper. *Observation* (Section V): Real-ESRGAN reduces mean NC to 0.795 while achieving SSIM of 0.906, the highest perceptual quality of any attack we test — a combination absent from every classical attack profile. ESPCN [7], passed through the identical upscale–downscale pipeline, leaves watermarks intact, isolating the cause to the learned reconstruction. *Mechanism* (Section VI): a stability analysis over 102,400 DCT blocks shows Real-ESRGAN concentrates damage in the DC coefficient at 5–14$\times$ the magnitude of any other position, whereas JPEG spreads damage uniformly — a perturbation profile under which the classical embedding positions, chosen for compression-era attacks, are no longer well placed. *Design implication and validation* (Section VI): if the damage profile is attack-specific, coefficient selection should be attack-aware. A sweep of 43 coefficient pairs confirms partial mitigation is achievable — positions $(7,5)/(7,7)$ improve Real-ESRGAN NC from 0.795 to 0.833 and eliminate imperceptibility failures — but at a quantified cost to JPEG robustness, with no configuration improving against all attacks simultaneously.

Our contributions are:

- To the best of our knowledge, the first systematic evaluation of a classical DWT-DCT scheme under learned super-resolution, demonstrating that AI enhancement exhibits a quality–robustness profile unreachable by classical attacks, with ESPCN serving as a pipeline control that attributes the effect to the learned reconstruction.
- A mechanistic characterization showing AI enhancement perturbs the DCT domain in a fundamentally different pattern than JPEG compression, which changes the assumptions behind classical coefficient-pair selection.
- An empirical evaluation of attack-aware coefficient selection as the design principle that follows, including its benefits, its costs, and negative results that delimit what

position selection can achieve.

Section II reviews related work. Section III summarizes the DWT-DCT scheme. Section IV describes the experimental setup. Sections V and VI present the observation, mechanism, and validation. Section VII concludes.

II. Related Work

A. Classical Watermarking and Robustness Evaluation

Transform-domain methods dominate invisible watermarking [2], embedding signals in DCT [8], DWT [9], or combined coefficients. Al-Haj [3] introduced the combined DWT-DCT scheme evaluated here. Subsequent variants add SVD [10] or optimization-based coefficient selection [11]. Across this lineage, robustness evaluation follows the protocol established by Petitcolas et al. [1]: compression, noise, filtering, and geometric attacks. We are not aware of prior work in this lineage that evaluates classical frequency-domain watermarks against AI-based image enhancement.

B. Deep Learning Watermarking

HiDDeN [12] introduced encoder–noise–decoder training; StegaStamp [13] extended it to physical-world robustness. Recent methods harden watermarks against generative edits: the W-Bench/VINE work [6] leverages diffusion priors, and SuperMark [14] uses diffusion-based super-resolution as an embedding tool. These methods show super-resolution models are powerful enough to serve as watermarking tools; our results show the complementary finding that the same model class threatens classical watermarks.

C. AI-Based Watermark Removal and Benchmarks

Zhao et al. [15] showed diffusion regeneration removes neural watermarks. WAVES [4] systematized robustness benchmarking across 26 attacks for neural schemes, evaluating DWT-DCT only briefly in an appendix and without enhancement-class attacks. W-Bench [6] broadened coverage to image and video editing, again for neural watermarks only. Neither benchmark evaluates classical frequency-domain watermarks against AI enhancement; the present work addresses this gap.

D. AI Image Enhancement

Real-ESRGAN [5] achieves blind restoration with purely synthetic training and is widely deployed in consumer applications. ESPCN [7] performs efficient sub-pixel convolution for real-time super-resolution. Unlike regeneration attacks, which discard and reconstruct content, enhancement models refine images in place while restructuring frequency content — a mechanistic distinction that motivates evaluating enhancement as a candidate attack category in its own right.

III. DWT-DCT Watermarking

A. Scheme Overview

The scheme is blind: extraction requires only the secret key, no reference image [2]. The host image is converted from RGB to YCbCr (ITU-R BT.601); embedding modifies only the luminance channel Y, reflecting the visual system's greater sensitivity to luminance distortion.

B. Embedding

A single-level Haar DWT applied to Y yields subbands LL, LH, HL, HH. Watermark bits are embedded in the LL subband, partitioned into non-overlapping 8×8 blocks; each block encodes one bit via the 2D DCT of the block. Each bit $b \in \{0, 1\}$ is encoded by enforcing a relative ordering between coefficients at positions (r_1, c_1) and (r_2, c_2) with margin δ:

$$b = 1 : C(r_1, c_1) \leftarrow \max\big(C(r_1, c_1),\, C(r_2, c_2) + \delta\big) \quad (1)$$

$$b = 0 : C(r_2, c_2) \leftarrow \max\big(C(r_2, c_2),\, C(r_1, c_1) + \delta\big) \quad (2)$$

Modified coefficients are inverse-DCT and inverse-DWT transformed and the image is converted back to RGB.

C. Extraction

The same transform pipeline is applied to the (possibly attacked) image and each bit recovered by comparison:

$$\hat{b} = \begin{cases} 1 & \text{if } C(r_1, c_1) > C(r_2, c_2) \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

D. Parameters and Metrics

Baseline positions are $(r_1, c_1) = (4, 1)$, $(r_2, c_2) = (1, 4)$ with $\delta = 30.0$; a 128-bit watermark is embedded with fixed seed 42 at four strength multipliers $\alpha \in \{0.05, 0.1, 0.2, 0.3\}$. Recovery is measured by normalized correlation between the original watermark w and extraction $\hat{w}$, both mapped to $\{-1, +1\}$:

$$\mathrm{NC} = \frac{w \cdot \hat{w}}{\|w\|\,\|\hat{w}\|} \quad (4)$$

Under the bipolar mapping, $\mathrm{NC} \in [-1, 1]$: 1.0 indicates perfect recovery and values near 0 indicate extraction at chance level.

IV. Experimental Setup

A. Dataset

100 images: 24 from the Kodak suite [16] and 76 from the TAMPERE17 noise-free database [17], all resized to 512×512 RGB.

B. Attack Conditions

Classical attacks. JPEG quality 50 (JPEG-Q50), Gaussian noise ($\sigma = 10$), 3×3 median filter, 5×5 Gaussian blur, and 80% crop with resize to 512×512.

AI enhancement attacks. Real-ESRGAN [5], a GAN-based blind super-resolution model, and ESPCN [7], a lightweight convolutional model. Both upscale watermarked images $4\times$ to 2048×2048 and resize back to 512×512 via Lanczos resampling, simulating a realistic user enhancement workflow. Because the two attacks share this identical upscale–downscale wrapper and differ only in the enhancement network, ESPCN additionally functions as a *pipeline control*: any difference in watermark damage between the two is attributable to the learned reconstruction, not to resampling.

TABLE I

MEAN WATERMARK SURVIVAL AND IMAGE QUALITY BY ATTACK, AVERAGED ACROSS 100 IMAGES AND FOUR EMBEDDING STRENGTHS. NC AND BER MEASURE WATERMARK RECOVERY; PSNR AND SSIM MEASURE PERCEPTUAL QUALITY AFTER ATTACK.

Attack	NC	BER	PSNR	SSIM
Watermarked only	0.999	0.001	48.0	0.999
Gaussian noise	0.999	0.001	28.1	0.809
JPEG-Q50	0.996	0.002	29.8	0.886
Gaussian blur	0.922	0.039	25.7	0.765
Median filter	0.892	0.054	26.6	0.785
Crop-Resize	−0.010	0.505	14.4	0.129
ESPCN 4×	0.999	0.000	–	–
Real-ESRGAN 4×	0.795	0.103	–	0.906

C. Metrics

NC, BER, PSNR, and SSIM [18] are computed on the Y channel per image per attack and averaged across the 100 images.

D. DCT Stability Analysis

We apply the DWT and 8×8 block DCT to each watermarked image and its Real-ESRGAN-enhanced counterpart at $\alpha = 0.1$, computing the mean absolute coefficient difference per DCT position across all blocks — 102,400 blocks in total — and repeat for JPEG-Q50 for direct comparison.

E. Frequency Sweep

43 candidate coefficient pairs are drawn from three groups: symmetric pairs, near-symmetric pairs (mirrored-coordinate Manhattan distance ≤ 2), and asymmetric pairs selected by stability differential. Each configuration is screened against Real-ESRGAN, JPEG-Q50, and Gaussian blur at $\alpha = 0.1$ on a 20-image subset; the top-performing pair is then re-evaluated on all 100 images at all four α values.

F. Implementation

PyTorch, PyWavelets, and OpenCV on Apple M2 hardware; Real-ESRGAN via the `py-real-esrgan` package, ESPCN via OpenCV DNN Super-Resolution (4×). Code is available at [REPO].

V. RESULTS: THE QUALITY–WATERMARK ASYMMETRY

A. Watermark Survival Under Attack

Table I reports mean NC, BER, PSNR, and SSIM. Classical signal-processing attacks leave watermarks largely intact: Gaussian noise and JPEG-Q50 produce NC above 0.99, consistent with prior DWT-DCT evaluations [3]; blur and median filtering cause moderate degradation (0.922, 0.892); Crop-Resize destroys the watermark (NC ≈ -0.01, i.e. chance-level extraction), consistent with the known vulnerability of block-aligned transform-domain schemes to geometric desynchronization.

The two enhancement attacks diverge sharply. ESPCN leaves watermarks indistinguishable from the unattacked baseline (NC $\approx$ 0.999). Real-ESRGAN reduces mean NC to 0.795 with BER 0.103 — more destructive than any classical attack except Crop-Resize — while simultaneously achieving SSIM of 0.906, the highest perceptual quality of any attack in our evaluation. Because both models share the identical resampling wrapper, this contrast indicates the damage arises from Real-ESRGAN's learned reconstruction itself, and that AI enhancement is not inherently threatening: the threat is architecture-specific.

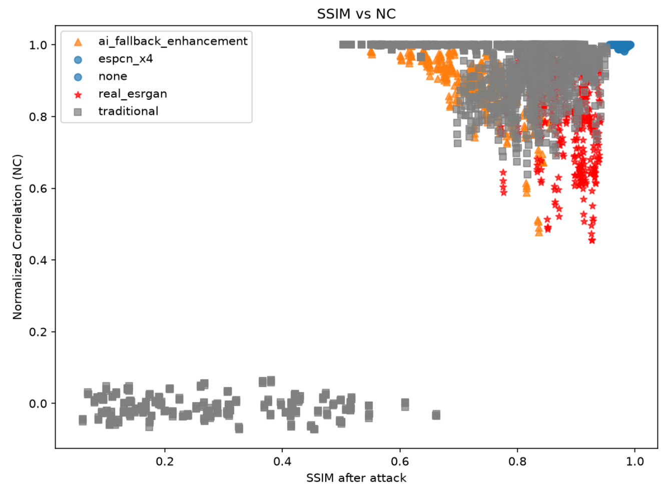

Fig. 1. SSIM vs. NC per attacked image. Real-ESRGAN occupies a high-SSIM, low-NC region unreachable by any classical attack; ESPCN clusters near NC = 1.0 at high SSIM. Classical attacks that reduce NC also reduce SSIM.

TABLE II

MEAN NC UNDER REAL-ESRGAN BY EMBEDDING STRENGTH α.

α	NC (Real-ESRGAN)	SSIM
0.05	0.787	0.907
0.10	0.797	0.906
0.20	0.815	0.906
0.30	0.832	0.905

B. The Asymmetry

Figure 1 shows the central observation. Every classical attack that produces low NC also produces low SSIM — watermark damage is coupled to visible quality damage. Real-ESRGAN decouples them, occupying a high-quality, low-detection region that no classical attack reaches. Operationally, a content owner monitoring watermark integrity would observe degraded detection with no visible indication that the image had been modified. This asymmetry, rather than the magnitude of NC loss alone, is what distinguishes enhancement from the classical attack repertoire.

C. Embedding Strength Does Not Restore Robustness

Table II shows NC improves only marginally as α quadruples from 0.05 to 0.30, while SSIM remains flat. For classical attacks, stronger embedding buys proportionally better survival; Real-ESRGAN does not exhibit this behavior. This is consistent with the model overwriting the DCT coefficient relationships used for bit encoding during reconstruction, rather

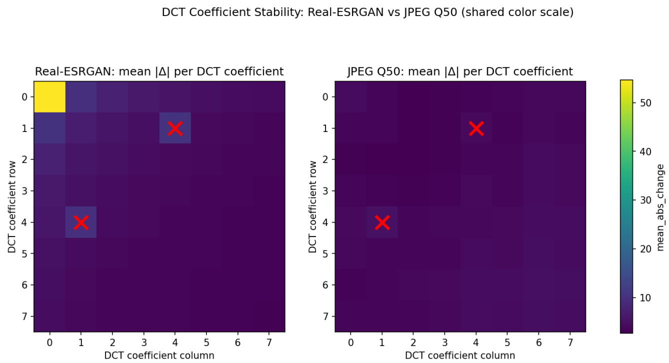

Fig. 2. Mean absolute DCT coefficient perturbation per position for Real-ESRGAN (left) and JPEG-Q50 (right) across 102,400 blocks, shared color scale. Real-ESRGAN concentrates damage in the DC coefficient; JPEG distributes damage nearly uniformly. Baseline embedding positions $(4, 1)$ and $(1, 4)$ are marked.

TABLE III
MEAN NC BY COEFFICIENT-PAIR SYMMETRY CATEGORY (20-IMAGE SCREENING SET, $\alpha = 0.1$).

Category	ESRGAN	JPEG	Blur	Overall
Symmetric	0.821	0.866	0.959	0.911
Near-symmetric	0.856	0.721	0.976	0.888
Asymmetric	0.689	0.868	0.735	0.822

than attenuating the watermark signal — an interpretation the next section examines directly.

VI. MECHANISM AND MITIGATION

A. Why: DCT Coefficient Damage Profiles

Figure 2 shows mean absolute perturbation per DCT position. Real-ESRGAN concentrates damage almost entirely in the DC coefficient at $(0, 0)$, with mean perturbation roughly 5 to 14 times greater than any other position, and elevated perturbation extending into the low- and mid-frequency region. This pattern is consistent with the learned reconstruction re-rendering local luminance statistics; notably, it cannot be attributed to the upscale–downscale resampling itself, since ESPCN shares that pipeline and leaves extraction nearly perfect. JPEG-Q50, by contrast, distributes damage across all 64 positions in a narrow range with mild elevation at higher frequencies, consistent with its quantization table design.

The design consequence is direct: the classical embedding positions $(4, 1)$ and $(1, 4)$ — well chosen under JPEG's uniform profile — sit in a mid-frequency zone where Real-ESRGAN's perturbation is elevated relative to the most stable high-frequency corner positions. The coefficient-selection assumptions inherited from the compression era do not transfer to the enhancement threat. If damage profiles are attack-specific, coefficient selection should be attack-aware; the remainder of this section evaluates that principle.

B. Validation: Frequency Sweep

Symmetry. Symmetric pairs achieve mean NC of 0.821 under Real-ESRGAN versus 0.689 for asymmetric pairs (Table III), disconfirming the hypothesis that coordinate symmetry contributes to the vulnerability.

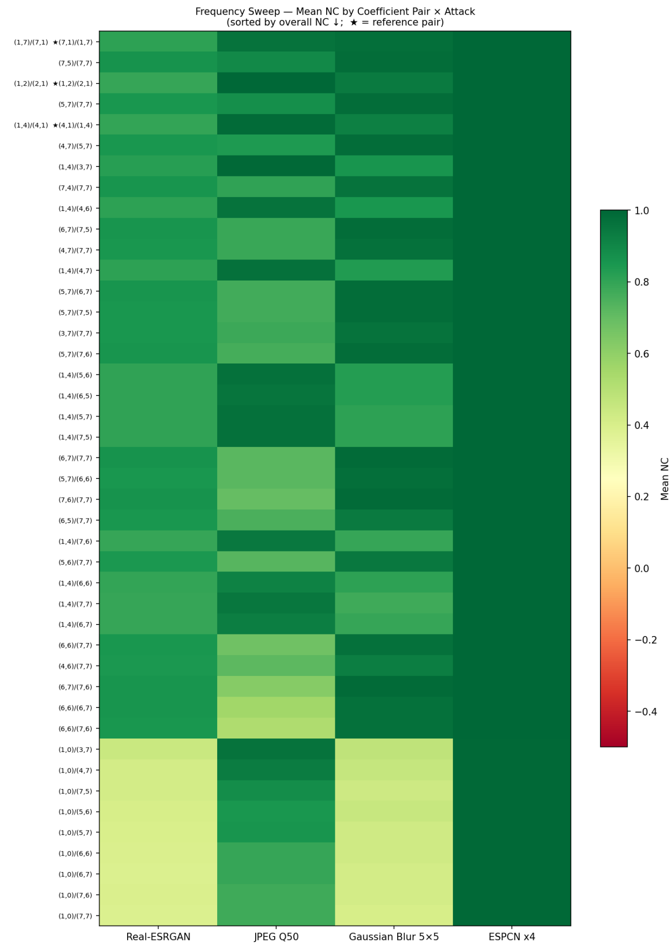

Fig. 3. Real-ESRGAN NC vs. JPEG-Q50 NC per coefficient pair. No pair dominates both axes simultaneously, consistent with a tradeoff between AI-enhancement and compression robustness.

High-frequency corner pairs. The top-performing pairs under Real-ESRGAN are anchored at the corner position $(7, 7)$, consistent with the damage profile of Figure 2: $(7, 5)/(7, 7)$ achieves $\text{NC} = 0.864$ on the screening set. Re-evaluated across all 100 images at all four α values, this pair improves Real-ESRGAN NC from 0.795 to 0.833, at the cost of JPEG regression from 0.996 to 0.903.

Imperceptibility. The repositioning also carries an unambiguous benefit: baseline positions $(4, 1)/(1, 4)$ produce a minimum per-image PSNR of 36.71 dB, below the conventional 40 dB imperceptibility threshold on some images, whereas $(7, 5)/(7, 7)$ achieves a minimum of 41.56 dB, maintaining imperceptibility across all tested images.

No universal improvement. No configuration improves against all three attacks simultaneously (Figure 3): pairs that improve Real-ESRGAN survival regress under compression. Embedding position selection provides partial, attack-aware mitigation, not a resolution of the vulnerability.

C. Subband Selection Does Not Help

As a supplementary comparison, we evaluated a DWT-SVD scheme embedding in the HH subband at matched imperceptibility (47.7 vs. 48.0 dB PSNR). It achieves NC of only 0.18 under Gaussian blur and JPEG-Q50 and 0.29 under Real-ESRGAN — near-chance extraction under attacks the DWT-DCT-LL scheme survives at 0.93, 0.98, and 0.75 respectively. Because this comparison changes both the transform (DCT to SVD) and the subband (LL to HH), it cannot isolate the subband's contribution; it does, however, suggest that moving embedding energy to high-frequency subbands is not by itself a viable escape from the enhancement vulnerability.

D. Limitations

Our evaluation covers two enhancement architectures; characterizing enhancement as a general attack category will require evidence across a broader model family, and our results themselves show the threat is architecture-specific. All results are reported as means across 100 images without variance estimates. The evaluation targets one classical scheme (DWT-DCT with ordering-based bit encoding); generalization to other classical schemes remains to be established. The frequency sweep screens candidates on a 20-image subset before full validation of the selected pair, so sweep-level comparisons (Table III) carry lower statistical weight than the headline results.

VII. Conclusion

We investigated whether AI-based image enhancement violates the quality–attack tradeoff underlying classical watermark robustness evaluation, and found that it can: Real-ESRGAN degrades DWT-DCT watermark recovery to $\mathrm{NC} = 0.795$ while producing the highest perceptual quality of any attack tested, a region no classical attack reaches, while ESPCN — sharing the identical resampling pipeline — leaves watermarks intact. The mechanism is a DC-concentrated perturbation profile fundamentally unlike JPEG's uniform profile, under which classically chosen embedding positions are no longer well placed. Attack-aware repositioning to $(7,5)/(7,7)$ recovers part of the loss and improves worst-case imperceptibility, but at a quantified cost to compression robustness; no position improves against all attacks simultaneously.

Content owners relying on DWT-DCT watermarking should be aware that consumer enhancement tools can silently degrade watermark integrity. We recommend that robustness evaluation frameworks adopt AI enhancement as a standard attack category. Future work should investigate content-adaptive strategies that jointly optimize coefficient selection and embedding strength against both classical and enhancement-class attacks, and should extend the evaluation across a broader family of enhancement models.

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